

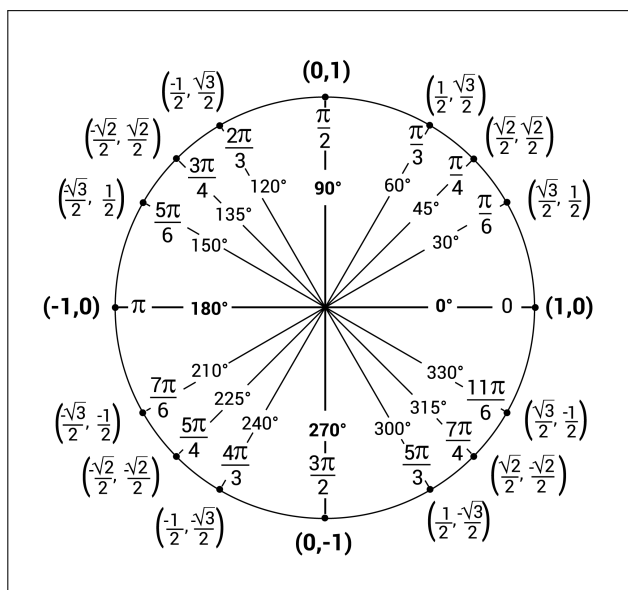
Extra Unit A practice 1

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity A and Jubilee 1.

Formulas

- $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1, \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$
- Continuity at $x = a$: $\lim_{x \rightarrow a} f(x) = f(a)$
- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- Basic derivatives
 - $\frac{d}{dx} (c \cdot f(x)) = c \cdot f'(x)$
 - $\frac{d}{dx} (f(x) \pm g(x)) = f'(x) \pm g'(x)$
 - $\frac{d}{dx} (x^n) = nx^{n-1}$
 - $\frac{d}{dx} (e^x) = e^x$
 - $\frac{d}{dx} (\ln(x)) = \frac{1}{x}$
 - $\frac{d}{dx} (\sin x) = \cos x$
 - $\frac{d}{dx} (\cos x) = -\sin x$
 - $\frac{d}{dx} (\tan x) = \sec^2 x$
 - $\frac{d}{dx} (\sec x) = \sec x \cdot \tan x$
 - $\frac{d}{dx} (\cot x) = -\csc^2 x$
 - $\frac{d}{dx} (\csc x) = -\csc x \cdot \cot x$

- Derivative rules
 - $\frac{d}{dx} (f(x) \cdot g(x)) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
 - $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
 - $\frac{d}{dx} (f(g(x))) = f'(g(x)) \cdot g'(x)$
- $L(x) = f'(a)(x - a) + f(a)$



A1 - Limits and continuity

(a) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sin(x) + 100e^{\cos(x)}}{x^2}$. *Hint: Use the Squeeze Theorem.*

$$\frac{-1 + 100e^{-1}}{x^2} \leq \frac{\sin(x) + 100e^{\cos(x)}}{x^2} \leq \frac{1 + 100e}{x^2}$$

$$\begin{array}{ccc} \downarrow x \rightarrow -\infty & & \downarrow x \rightarrow -\infty \\ 0 \leq \lim_{x \rightarrow -\infty} \frac{\sin(x) + 100e^{\cos(x)}}{x^2} \leq 0 \end{array}$$

|| ← By Squeeze Theorem
0

(b) Consider the function

$$f(x) = \begin{cases} \frac{5x}{\sqrt{mx+b}-9} & x \neq 0 \\ 15 & x = 0 \end{cases}$$

Find values of m and b that make $f(x)$ continuous at $x = 0$.

$$\lim_{x \rightarrow 0} \frac{5x}{\sqrt{mx+b}-9} \cdot \frac{\sqrt{mx+b}+9}{\sqrt{mx+b}+9} = \lim_{x \rightarrow 0} \frac{5x(\sqrt{mx+b}+9)}{mx+b-81} \rightarrow \boxed{b=81}$$

otherwise the limit is 0

$$= \lim_{x \rightarrow 0} \frac{5}{m} (\sqrt{mx+81}+9) = \frac{90}{m} = 15 \rightarrow \boxed{m=6}$$

A2 - The derivative

- (a) Use the limit definition of the derivative to show that if $f(x) = x^4$, then $f'(x) = 4x^3$. It will be helpful to have the expansion $(x+h)^4 = x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4$.

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h} &= \lim_{h \rightarrow 0} \frac{x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4 - x^4}{h} \\ &= \lim_{h \rightarrow 0} 4x^3 + 6x^2h + 4xh^2 + h^3 = 4x^3\end{aligned}$$

- (b) Let $g(x) = |x|$. Use the limit definition of the derivative to find $g'(0)$ or show that $g'(0)$ doesn't exist.

$$g'(0) = \lim_{h \rightarrow 0} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$

$$\lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

$$\lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1 \rightarrow \neq \text{ so } g'(0) \text{ DNE}$$

A3 - Derivative rules

(a) Consider the function

$$g(x) = \ln(\ln(-4x + 4))$$

Compute $g'(x)$.

$$g'(x) = \frac{1}{\ln(-4x+4)} \cdot \frac{1}{-4x+4} \cdot -4$$

(b) Find the derivative of $f(x) = x^{\frac{3x^2+1}{\tan(x)}}$. You do not need to simplify your answer.

$$f(x) = e^{\frac{3x^2+1}{\tan(x)} \cdot \ln(x)}$$

$$f'(x) = e^{\frac{3x^2+1}{\tan(x)} \cdot \ln(x)} \cdot \left(\frac{3x^2+1}{\tan(x)} \cdot \frac{1}{x} + \frac{\tan x \cdot 6x - (3x^2+1)\sec^2 x}{\tan^2 x} \cdot \ln x \right)$$

A4 - Implicit curves and linearization

(a) Consider the curve given by the equation

$$\sqrt{xy} = \frac{x^2}{y^3} - 4$$

Find $\frac{dy}{dx}$ at $(x, y) = (8, 2)$. You do not need to simplify your answer.

$$\frac{1}{2}(xy)^{-\frac{1}{2}} \cdot \left(1 \cdot y + x \cdot \frac{dy}{dx}\right) = \frac{y^{\cancel{2}} \cdot 2x - x^2 \cdot 3y^{\cancel{2}} \cdot \frac{dy}{dx}}{y^{\cancel{6}4}}$$

$$\frac{1}{2}(16)^{-\frac{1}{2}} \left(2 + 8 \frac{dy}{dx}\right) = \frac{2 \cdot 16 - 64 \cdot 3 \frac{dy}{dx}}{16} = 2 - 12 \frac{dy}{dx}$$

$$\frac{1}{4} + \frac{dy}{dx} = 2 - 12 \frac{dy}{dx}$$

$$13 \frac{dy}{dx} = \frac{7}{4}$$

$$\frac{dy}{dx} = \frac{7}{52}$$

(b) Consider the following function $f(x)$ along with its derivative $f'(x)$:

$$f(x) = \frac{(1-x)^2}{1+(1+x)^2}$$

$$f'(x) = \frac{4x^2 + 2x - 6}{(x^2 + 2x + 2)^2}$$

Find the linear approximation of $f(x)$ at the value $a = 0$, and then use the linear approximation to estimate

$$\frac{0.99^2}{1 + 1.01^2}$$

You do not need to simplify your answer.

$$f(0) = \frac{(1-0)^2}{1+(1+0)^2} = \frac{1}{2}$$

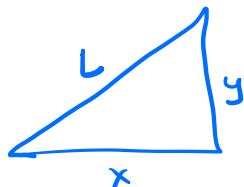
$$f'(0) = \frac{-6}{2^2} = -\frac{3}{2}$$

$$L(x) = -\frac{3}{2}(x-0) + \frac{1}{2}$$

$$\frac{0.99^2}{1 + 1.01^2} = f(0.01) \approx L(0.01) = \boxed{-\frac{3}{2}(0.01-0) + \frac{1}{2}}$$

A5 - Related rates

- (a) One leg of a right triangle is 3 cm long and increasing at a rate of 6 cm/s, while the other leg is 4 cm long and decreasing at a rate of 2 cm/s. What is the rate of change of the hypotenuse's length?



$$x^2 + y^2 = L^2$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 2L \cdot \frac{dL}{dt}$$

$$x = 3 \quad \frac{dx}{dt} = 6$$

$$2 \cdot 3 \cdot 6 + 2 \cdot 4 \cdot (-2) = 2 \cdot 5 \cdot \frac{dL}{dt}$$

$$y = 4 \quad \frac{dy}{dt} = -2$$

$$\rightarrow \frac{dL}{dt} = 2 \text{ cm/s}$$

$$L = \sqrt{3^2 + 4^2} = 5$$

- (b) The relationship between the pressure, P (in kPa), and volume, V (in cubic centimeters), of an expanding gas is given by the equation

$$PV^{1.4} = 384$$

If at some instant the pressure of the gas is 3 kPa and decreasing at a rate of 21 kPa per minute, what is the rate of change of the volume? Include units in your answer.

$$\frac{dP}{dt} \cdot V^{1.4} + P \cdot 1.4 V^{0.4} \cdot \frac{dV}{dt} = 0$$

$$P = 3 \rightarrow 3 \cdot V^{1.4} = 384$$

$$V^{\frac{7}{5}} = 128$$

$$-21 \cdot 32^{1.4} + 3 \cdot 1.4 \cdot 32^{0.4} \cdot \frac{dV}{dt} = 0$$

$$V = 128^{\frac{5}{7}} = 32$$

$$\frac{dV}{dt} = \frac{21 \cdot 128}{3 \cdot 1.4 \cdot 4} = \boxed{160 \text{ cm}^3/\text{min}}$$